Name

Mathematics Advanced Year 12

# Continuous Random Variables

**Book 1**

Version: 261021

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# Syllabus Content

**MAV-12-07** solves problems involving discrete probability distributions, continuous random variables and the normal distribution

**Continuous random variables**

- Estimate the probability that a continuous random variable falls in some interval using relative frequencies and histograms obtained from data
- Recognise that the probability of a particular value for a continuous random variable is 0 and hence that $P(a < X < b) = P(a \leq X < b) = P(a < X \leq b) = P(a \leq X \leq b)$ since $P(X = a) = P(X = b) = 0$ when $X$ is a continuous random variable
- Define the cumulative distribution function (CDF), $F(x)$, as the probability of a random variable, $X$, having values less than or equal to $x$, so $F(x) = P(X \leq x)$ and $P(a \leq x \leq b) = F(b) - F(a)$
- Recognise that the cumulative distribution function, $F(x)$, is non-decreasing for all $x$ in its domain, and graph cumulative distribution functions, given a formula for $F(x)$, with and without graphing applications
- Define a probability density function (PDF), $f(x)$, for a random variable $X$ with cumulative distribution function $F(x)$ as $f(x) = F'(x)$ and recognise that $P(a < X < b) = \int_a^b f(x)\,dx$
- Recognise the properties of a probability density function: $f(x) \geq 0$ for all $x$ in the domain of $f(x)$, and $\int_a^b f(x)dx = 1$ if the domain of $f(x)$ is $[a, b]$, or $\int_{-\infty}^{\infty} f(x)dx = 1$ if the domain of $f(x)$ is all real $x$
- Apply the properties of a probability density function to solve problems and justify conclusions
- Find the mode from a given probability density function
- Obtain the cumulative distribution function using the formula $F(x) = \int_a^x f(t)dt$ where $f(x)$ is a given probability density function defined on the interval $[a, b]$
- Determine and use the probability density function $f(x) = \frac{1}{b-a}$ for a continuous uniform distribution for a random variable $X$ taking values in the interval $[a, b]$
- Use a cumulative distribution function to calculate the median and quartiles for a continuous random variable
- Find the probability density function from a given cumulative distribution function
- Generate the expression for the expected value of a continuous random variable, $E(X) = \mu = \int_a^b xf(x)dx$, where the probability density function $f(x)$ is defined on the interval $[a, b]$
- Generate the expression for the variance of a continuous random variable, $\text{Var}(X) = E(X^2) - \mu^2 = \int_a^b x^2 f(x)dx - \mu^2$, where the probability density function $f(x)$ is defined on the interval $[a, b]$
- Evaluate the expected value and the variance of a continuous random variable, where the probability density function $f(x)$ is defined on the interval $[a, b]$, that involve integration of functions within the scope of the Mathematics Advanced course
- Evaluate the expected value and the variance of a continuous random variable, where the probability density function $f(x)$ is defined on the interval $[a, b]$, that involve integration of functions beyond the scope of the Mathematics Advanced course using an online computational application

# Continuous Random Variables and Relative Frequency

## Review

☐ Analyse discrete random variables

| $x$ | 0 | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|---|
| $P(X = x)$ | 0.1 | 0.2 | 0.3 | 0.25 | 0.05 | 0.1 |

**a** $P(X = 1)$ 0.2

**b** $P(2 \leq X \leq 4)$ 0.6

**c** $P(1 \leq X < 4)$ 0.75

**d** $P(X = 6)$ 0

**e** $P(X \leq 2)$ 0.6

**f** $P(X < 4)$ 0.85

☐ Analyse grouped data

The table shows the times taken, in minutes, to complete a puzzle.

| Time taken, $t$ (minutes) | Frequency |
|---|---|
| $2 < t \leq 4$ | 16 |
| $4 < t \leq 6$ | 26 |
| $6 < t \leq 8$ | 38 |
| $8 < t \leq 10$ | 40 |

**a** Which class interval has the lowest frequency?

$2 < t \leq 4$

**b** How many people took longer than 8 minutes to complete the puzzle?

40

**c** How many people in total completed the puzzle?

120

**d** What is the relative frequency for people taking $6 < t \leq 10$ to complete the puzzle?

13/20

☐ Recall the law of large numbers

The Law of Large Numbers says that as you …… the number of trials, the relative frequency gets closer to the t…… p……

## ⓘ Continuous Random Variables

A **continuous** random variable can take any value in an interval, not just specific isolated values. Examples include height, mass, temperature, and waiting time.

## ⓘ **Identify** Continuous and discrete random variables

| | | | |
|---|---|---|---|
| **a** | Number of cars in a car park<br><br>☐ Discrete<br>☐ Continuous | **b** | Speed of a car as it passes a sensor<br><br>☐ Discrete<br>☐ Continuous |
| **c** | Temperature at noon on a given day<br><br>☐ Discrete<br>☐ Continuous | **d** | Number of defective items in a batch<br><br>☐ Discrete<br>☐ Continuous |
| **e** | Height of a student, in centimetres<br><br>☐ Discrete<br>☐ Continuous | **f** | Number of students absent on a day<br><br>☐ Discrete<br>☐ Continuous |
| **g** | Time for a Year 7 to swim 50 m<br><br>☐ Discrete<br>☐ Continuous | **h** | Score in a 10-question multi-choice quiz<br><br>☐ Discrete<br>☐ Continuous |
| **i** | Mass of a bag of rice<br><br>☐ Discrete<br>☐ Continuous | **j** | Number of heads when flipping 3 coins<br><br>☐ Discrete<br>☐ Continuous |

### ⓘ $P(X = a) = 0$ for Continuous Random Variables

For a continuous random variable, the **probability of any single exact value is 0**, because infinitely many values are possible in any interval.

In other words, $\boldsymbol{P(X = a) = 0}$ for all $a$.

The probability that a student in the school is *exactly* 158.00000… cm is zero.

You can only sensibly ask questions like 'what is the probability of someone's height being between 150 and 160 cm; $P(150 < X < 160)$?'

Furthermore, because $P(X = a) = 0$ and $P(X = b) = 0$, **including or excluding endpoints makes no difference** when calculating the probability of a value lying within an interval.

$$P(a < X < b) = P(a \leq X \leq b)$$

### Example Estimate relative frequencies from class intervals

The waiting times (minutes) for 40 patients at a clinic are summarised below:

| Time | Freq | Relative Freq |
|---|---|---|
| $0 \leq t < 3$ | 4 | 0.10 |
| $3 \leq t < 6$ | 12 | 0.30 |
| $6 \leq t < 9$ | 16 | 0.40 |
| $9 \leq t < 12$ | 6 | 0.15 |
| $12 \leq t < 15$ | 2 | 0.05 |

**a** Verify the relative frequencies sum to 1.

$$0.1 + 0.3 + 0.4 + 0.15 + 0.05 = 1$$

**b** Estimate $P(X < 6)$.

$$\begin{aligned} P(X < 6) &= 0.1 + 0.3 \\ &= 0.4 \end{aligned}$$

**c** Estimate $P(3 \leq X < 12)$.

$$\begin{aligned} P(3 \leq X < 12) &= 0.3 + 0.4 + 0.15 \\ &= 0.85 \end{aligned}$$

### Guided Practice

**a** Using the same table above, find $P(X < 9)$

0.8

**b** Using the same table above, find $P(3 \leq X < 15)$

0.9

A new sample has classes with relative frequencies

[0, 4), [4, 8), [8, 12), [12, 16)
0.08, 0.42, 0.36, 0.14.

**c** Find $P(X < 8)$

0.5

**d** Find $P(4 \leq X < 12)$

0.78

**1** State whether each random variable is discrete or continuous.

**a** The size of T-shirts worn by people

Discrete

**b** The speed of cars as they pass a certain point

continuous

**c** The volume of water in a dam

continuous

**d** The number of seats on an aeroplane

discrete

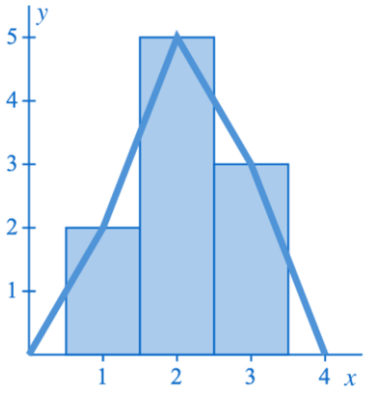
**e** The weight of babies born in August

Continuous

**2** A simple experiment generated the following data:

**a** Construct a frequency histogram and frequency polygon (join centres of data points back to horizontal axis).

| Score $x$ | 1 | 2 | 3 |
|---|---|---|---|
| Frequency $f$ | 2 | 5 | 3 |




**b** Calculate the total area of the histogram rectangles.

Total area = $1(2) + 1(5) + 1(3) = 10$

**c** Calculate the area under the frequency polygon.

Area under polygon = 10 (triangles cut above fit into gaps below)

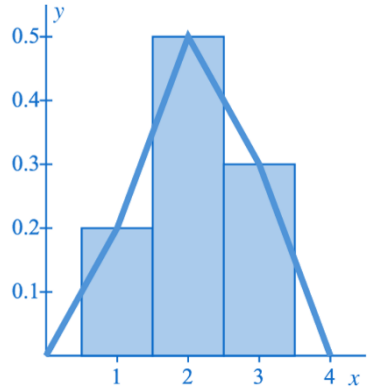
**d** What do you notice?

Both areas equal the total frequency.

**e** Add a row of relative frequencies (divide by total $n = 10$).

| Score $x$ | 1 | 2 | 3 |
|---|---|---|---|
| Frequency $f$ | 2 | 5 | 3 |
| Rel. Freq | | | |

0.2,0.5,0.3




**f** Construct a relative frequency histogram and polygon.

**g** Calculate the total area of the histogram rectangles.

Total area $= 1(0.2) + 1(0.5) + 1(0.3) = 1$

**h** Calculate the area under the relative frequency polygon.

Area under polygon = 1

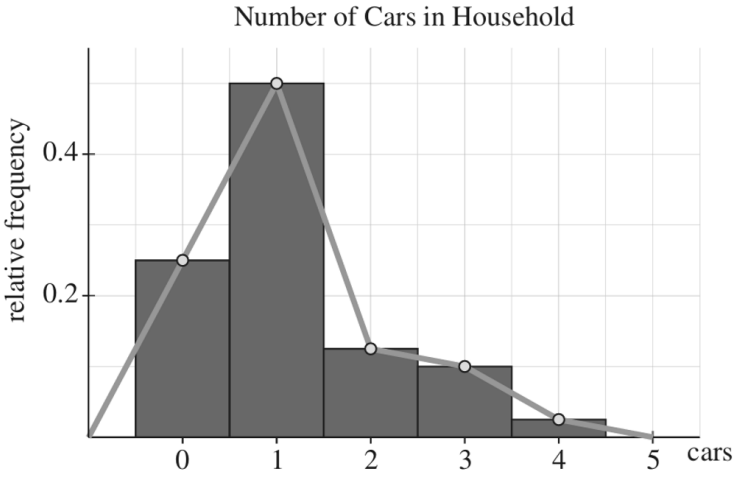
**i** What do you notice?

Both areas are the sum of the relative frequencies, 1

**j** What is the relationship between relative frequencies and the probabilities $P(X = x)$, according to the Law of Large Numbers?

With a large enough dataset, the relative frequency is an accurate experimental estimate of the theoretical probability.

**3** The number of cars owned per household in a suburb is recorded from census data. Results are shown in a relative frequency histogram.
(Relative frequencies equal probabilities since this is a population.)




**a** What fraction of households have no cars?

$P(0 \text{ cars}) = \frac{1}{4}$

**b** What fraction of households have fewer than 2 cars?

$P(\text{fewer than 2}) = P(0) + P(1) = \frac{3}{4}$

**c** What is the probability that a randomly selected household has three cars?

$P(3 \text{ cars}) = 0.1$

**d** Planners advise additional on-street parking if more than 40% of households have 3 or more cars. Will they advise additional parking?

$P(3 \text{ or more}) = 0.10 + 0.025 = 0.125 = 12.5\% < 40\%.$

Planners will not advise additional parking.

**e** Complete the probability distribution table $P(X = x)$ using the histogram.

| $x$ | 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| $P(X = x)$ | | | | | |

$0.25,\ 0.50,\ 0.125,\ 0.10,\ 0.025$

**f** Show the sum of probabilities is 1. How does this relate to the histogram area?

Sum = $0.25 + 0.5 + 0.125 + 0.1 + 0.025 = 1.$

The area of each rectangle = width × height = 1 × relative frequency = probability.

Total area = sum of probabilities = 1.

**g** Explain why the area under the relative frequency polygon equals the area of the histogram.

Each triangle above the polygon (cut off from a bar) fits into the gap below the polygon on the other side of that same bar's edge.

Total area is unchanged.

**h** Show the mean number of cars per household is 1.15.

$E(X) = 0(0.25) + 1(0.5) + 2(0.125) + 3(0.1) + 4(0.025) = 0.5 + 0.25 + 0.3 + 0.1 = 1.15.$

This is a population average: for a large number of households, the total cars divided by number of households approaches 1.15.

**i** For a street of 100 households, how many cars in total would you expect? Are the assumptions reasonable?

Expected cars = $100 \times 1.15 = 115$.

Assumption: the street's car-ownership distribution matches the suburb.

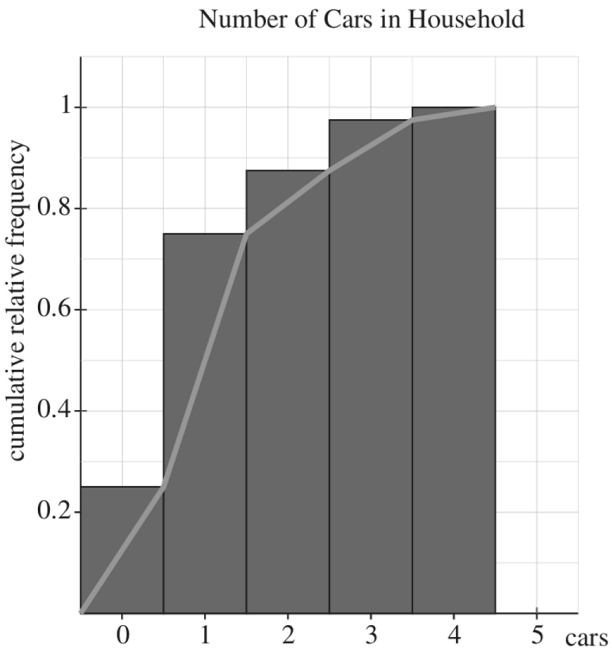
This may not hold, streets near train stations or in wealthier areas may differ.

**j** Complete the cumulative probability table $P(X \leq x)$.

| $x$ | 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| $P(X \leq x)$ | | | | | |

0.25, 0.75, 0.875, 0.975, 1

**k** Using horizontal lines at 0.25, 0.5, 0.75, find quartiles $Q_1$, $Q_2$, $Q_3$.




$Q_1$ at $P = 0.25$ gives $x \approx 0.5$,

$Q_2$ at $P = 0.5$ gives $x = 1$,

$Q_3$ at $P = 0.75$ gives $x \approx 1.5$.

# Probability Density Functions

## Review

☐ Integrate polynomials

**a**

$$\int_2^5 4x + 1\,dx$$

**b**

$$\int_0^4 \frac{3}{12}(4x + 1)\,dx$$

☐ Identify sign of a function

Find the domain for which $f(x) > 0$

**a** $f(x) = 2 - x$

**b** $f(x) = x^2 - 9$

**c** $f(x) = \frac{x}{8}$

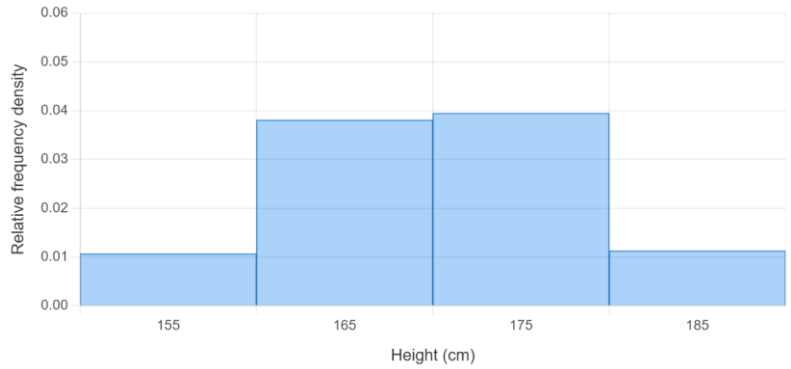
**d** $f(x) = \frac{3x^2}{32}$

## ⓘ Probability Density Functions

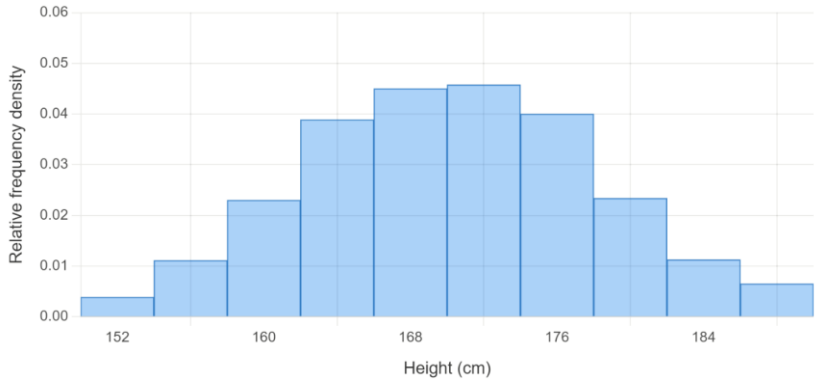
A **probability density function (PDF)** is a smooth curve that models the distribution of a continuous random variable. You can think of it as a relative frequency histogram where the class widths have been made infinitely narrow.

For example, imagine we measured the heights of every year 12 girl in the state:

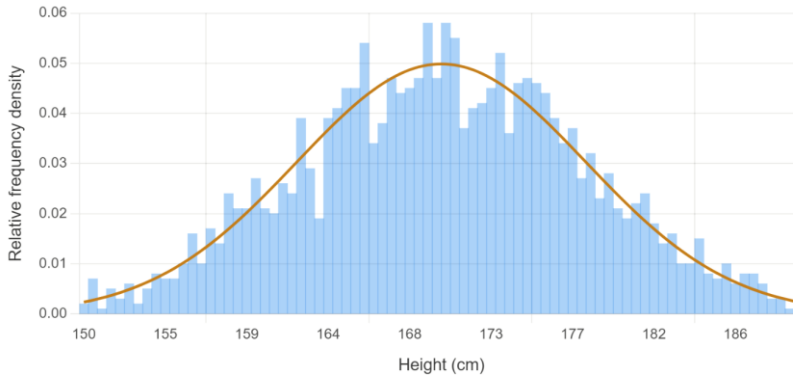
**Relative frequency histogram, range 150 – 190 cm, grouped in 4 classes of 10 cm width.**




**10 classes of 4 cm width:**




**80 classes of 0.5 cm width:**




The **area under the curve** between any two values equals the probability that the variable falls in that interval.

## Verifying Probability Density Functions

A function $f(x)$ is a PDF in the domain $[a, b]$ if and only if:

1. $f(x) \geq 0$ for all $x$ in $[a, b]$
2. $\int_a^b f(x)\,dx = 1$

## Example Determine whether function is a probability density function

Determine whether the function is a valid probability density function.

$$f(x) = \begin{cases} \frac{x}{8} & 0 \leq x \leq 4 \\ 0 & \text{otherwise} \end{cases}$$

1. $f(x) \geq 0$ for all $x$ in $[a, b]$:

$\frac{x}{8} \geq 0$ for $x \geq 0$,

$\therefore f(x) = \frac{x}{8} \geq 0$ for $0 \leq x \leq 4$

2. $\int_a^b f(x)\,dx = 1$

$\int_0^4 \frac{x}{8} dx = \left[\frac{x^2}{16}\right]_0^4 = \frac{16}{16} - 0 = 1$

$\therefore$ area under the curve in given region is 1.

In $[0,4]$, $f(x)$ is always positive and its integral is 1. Therefore, it is a PDF.

## Guided Practice

Determine whether these functions are valid PDFs

**a**

$f(x) = \frac{x}{18}$ on $[0,6]$

yes

**b**

$f(x) = \frac{3x^2}{26}$ on $[1,3]$

yes

**c**

$f(x) = x^2 - 3$ on $[0,3]$

No, not always positive

**d**

$f(x) = x - 1$ on $[1,3]$

No, integral not 1

## Key Ideas

1. A PDF must satisfy two conditions: $f(x) \geq 0$ for all $x$ in the domain, and the integral over the domain equals ……

**Example** Solve problems involving finding a constant in a PDF

A continuous probability distribution function has the function $f(x) = ax^2$ defined on $[1, 4]$.

Find the value of $a$.

$$\int_1^4 ax^2\,dx = 1$$

$$a\left[\frac{x^3}{3}\right]_1^4 = 1$$

$$a\left(\frac{64}{3} - \frac{1}{3}\right) = 1$$

$$21a = 1$$

$$a = \frac{1}{21}$$

## Guided Practice

Find the value of the constant if $f(x)$ is a valid PDF in the given domain.

**a** $f(x) = ax^2$ on $[0, 3]$

$\frac{1}{9}$

**b** $f(x) = ax^3$ on $[1, 4]$

$\frac{4}{255}$

**c** $f(x) = kx$ on $[2, 6]$

$\frac{1}{16}$

**d** $f(x) = ae^x$ on $[0, 2]$

$\frac{1}{e^2 - 1}$

## Key Ideas

1. The area under a probability density function must sum to ……, this means the integral over the domain must also equal ……
2. To find an unknown constant in a PDF, set the integral over the domain to equal ……

**1** Which of the following functions describe continuous probability distributions?

**a** $f(x) = 0.2$ in the domain $[1, 6]$

$f(x)$ always positive

$\int_1^6 0.2\ dx = 0.2(5) = 1, f(x) \geq 0$

Valid PDF.

**b** $f(x) = \frac{x}{12}$ in the domain $[0, 6]$

$f(x)$ always positive when $x > 0$

$\int_0^6 \left(\frac{x}{12}\right) dx = \left[\frac{x^2}{24}\right]_0^6 = \frac{3}{2} \neq 1$

Not a PDF.

**c** $f(x) = \frac{x^3}{324}$ for $0 \leq x \leq 3$

$f(x)$ always positive for $x > 0$

$\int_0^3 \left(\frac{x^3}{324}\right) dx = \left[\frac{x^4}{1296}\right]_0^3 = \frac{81}{1296} = \frac{1}{16} \neq 1$

Not a PDF.

**d** $f(x) = \frac{x^2}{21}$ in the interval $1 \leq x \leq 4$

$f(x)$ always positive

$\int_1^4 \left(\frac{x^2}{21}\right) dx = \left[\frac{x^3}{63}\right]_1^4 = \frac{63}{63} = 1, f(x) \geq 0$

Valid PDF.

**e** $f(x) = \frac{x}{8}$ for $1 \leq x \leq 8$

$f(x)$ always positive for $x > 0$

$\int_1^8 \left(\frac{x}{8}\right) dx = \left[\frac{x^2}{16}\right]_1^8 = \frac{63}{16} \neq 1$

Not a PDF.

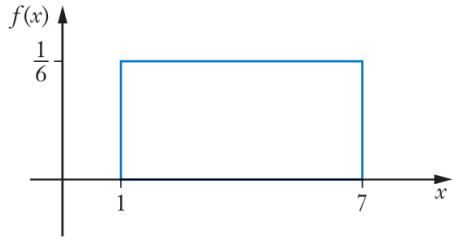
**4** Given the continuous probability distribution $f(x) = kx^3$ for $0 \leq x \leq 5$, 0 for all other $x$, find the value of $k$.

$\int_0^5 (kx^3)\ dx = k\left[\frac{x^4}{4}\right]_0^5 = \frac{625k}{4} = 1$

$k = \frac{4}{625}$

**5** Which of the following graphs are probability density functions?

**a**




Area $= \left(\frac{1}{6}\right)(6) = 1, f(x) \geq 0$

Valid PDF.

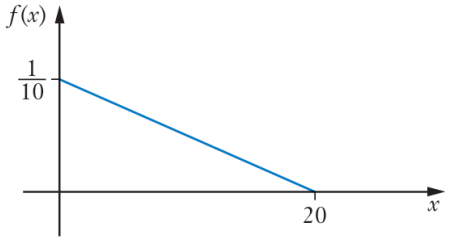
**b**




Area $= \left(\frac{1}{10}\right)(12) = \frac{12}{10} \neq 1$

Not a PDF.

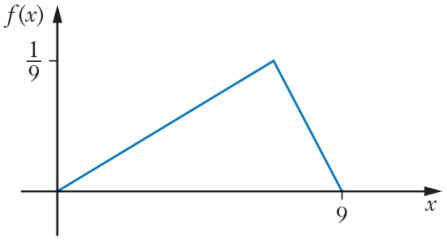
**c**




Area $= \left(\frac{1}{2}\right)(20)\left(\frac{1}{10}\right) = 1, f(x) \geq 0$

Valid PDF.

**d**




Area $= \left(\frac{1}{2}\right)(9)\left(\frac{1}{9}\right) = \frac{1}{2} \neq 1$

Not a PDF..

**6** Find $c$, given $f(x)$ is a PDF on the given domain.

**a** $y = cx^4$, domain $[0, 3]$

$$\int_0^3 c\,x^4\,dx = \frac{243c}{5} = 1$$

$$c = \frac{5}{243}$$

**b** $y = c$, domain $[0, 6]$

$$6c = 1$$

$$c = \frac{1}{6}$$

**c** $y = c$, domain $[-5, 5]$

$$10c = 1$$

$$c = \frac{1}{10}$$

**7** Find $c$, given $f(x)$ is a PDF on the given domain.

**a** $f(x) = \frac{8}{3}(1 - x)$, domain $[0, c]$

$$\frac{8}{3}\int_0^c (1 - x)\,dx = \frac{8}{3}\left(c - \frac{c^2}{2}\right) = 1$$

$$4c^2 - 8c + 3 = 0$$

$$(2c - 1)(2c - 3) = 0.\ c = \frac{1}{2} \text{ or } c = \frac{3}{2}$$

For $f(x) \geq 0$, need $1 - x \geq 0$ throughout, so $c \leq 1$.

$$c = \frac{1}{2}$$

**8** A function is given by $f(x) = \frac{x^2}{72}$. Over what domain starting at $x = 0$ is this a PDF?

$$\int_0^b \left(\frac{x^2}{72}\right) dx = \left[\frac{x^3}{216}\right]_0^b = \frac{b^3}{216} = 1$$

$$b^3 = 216, b = 6. \text{Domain } [0, 6]$$

**9** A PDF is given by $f(x) = \frac{2x^5}{87381}$ over the interval $1 \leq x \leq b$. Find the value of $b$.

$$\int_1^b \left(\frac{2x^5}{87381}\right) dx = \left(\frac{2}{87381}\right)\left[\frac{x^6}{6}\right]_1^b = \frac{b^6 - 1}{262143} = 1$$

$$b^6 = 262144 = 8^6, b = 8$$

**10** A PDF is given by $f(x) = ae^x$ over a certain domain. Find the exact value of $a$ if the domain is:

**a** $[1, 7]$

**b** $[0, 4]$

a) $a(e^7 - e) = 1$

$$a = \frac{1}{e(e^6 - 1)}$$

b) $a(e^4 - 1) = 1$

$$a = \frac{1}{e^4 - 1}$$

**11** Let $f(x) = \frac{3}{4}(x^2 - 4x + 3)$ be defined on [0, 4].

**a** Show that $\int_0^4 f(x)\, dx = 1$.

$$\int_0^4 \frac{3}{4}(x^2 - 4x + 3)\, dx = \frac{3}{4}\left[\frac{x^3}{3} - 2x^2 + 3x\right]_0^4$$

$$= \frac{3}{4}\left(\frac{64}{3} - 32 + 12\right)$$

$$= \frac{3}{4}\left(\frac{4}{3}\right) = 1$$

**b** Show that $f(x)$ is not a valid PDF.

$f(x) = \frac{3}{4}(x - 1)(x - 3)$.

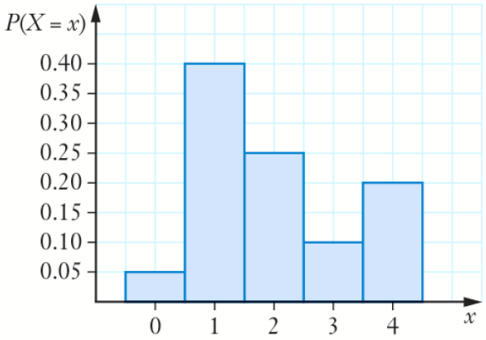
For $1 < x < 3$, $(x - 1) > 0$ and $(x - 3) < 0$, so $f(x) < 0$.

Not a valid PDF as $f(x)$ is not always positive.

# Calculating Probabilities using Integration

## Review

☐ Estimate probability in an interval using the relative frequency of classes




**a** Find $P(X = 1)$

**b** Find $P(X \leq 1)$

**c** $P(X \geq 2)$

☐ Work with area of a rectangle

The area of these rectangles is 1. Find the missing length.

**a**

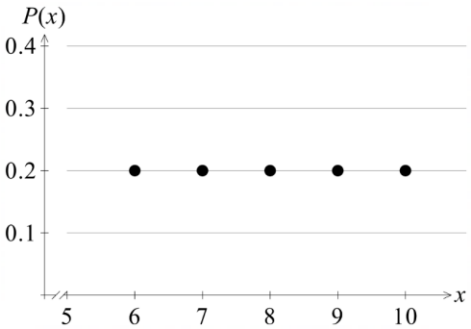



**b**




☐ Recall uniform distribution properties.

Consider the uniform distribution below. What is meant by a uniform distribution?




A probability distribution is uniform if the probability of each outcome is ……

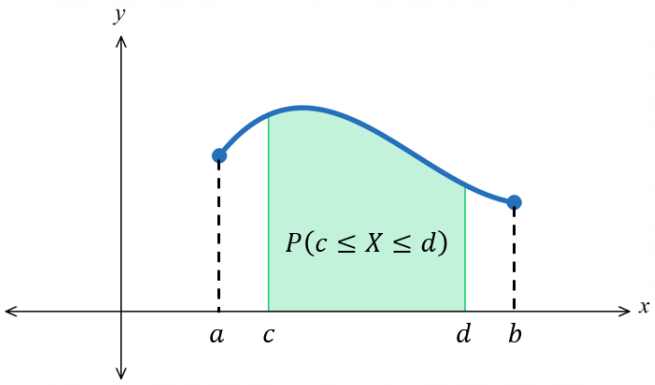
☐ Recall PDF properties

The area under a PDF must sum to ……

## ⓘ The Probability is the Definite Integral

For a probability density function (PDF) $f(x)$ defined on $[a, b]$, the probability that $X$ falls between $c$ and $d$ is:

$$P(c \leq X \leq d) = \int_c^d f(x)\ dx, \quad a \leq c \leq d \leq b$$




If the requested interval extends beyond the domain, the limits of integration are restricted to the domain boundary. For a PDF defined on [1, 4]:

$P(X \leq 3)$ integrates from 1 to 3, not from $-\infty$ or 0.

## ⓘ **Identify** integral from probability notation

Consider a PDF $f(x)$ defined for the domain $[1, 4]$. Write the integral for:

| | | |
|---|---|---|
| **a** $P(2 \leq X \leq 3)$ $\int_2^3 f(x)\,dx$ | **b** $P(0 \leq X \leq 3)$ $\int_1^3 f(x)\,dx$ | **c** $P(2 \leq X \leq 5)$ |
| **d** $P(X \leq 3)$ | **e** $P(X \geq 2)$ | **f** $P(-1 \leq X \leq 6)$ |
| **g** $P(1.5 \leq X \leq 4)$ | **h** $P(X \leq 5)$ | **i** $P(X \geq 0)$ |

## Example Calculate probability of a PDF by integration

The continuous random variable $X$ has PDF

$$f(x) = \frac{x^2}{21} \text{ on } [1, 4]$$

Find $P(2 \leq X \leq 3)$

$$\int_2^3 \frac{x^2}{21} dx$$

$$= \left[\frac{x^3}{63}\right]_2^3$$

$$= \frac{27}{63} - \frac{8}{63}$$

$$= \frac{19}{63}$$

Find $P(X \geq 3)$

$$\int_3^4 \frac{x^2}{21} dx$$

$$= \left[\frac{x^3}{63}\right]_3^4$$

$$= \frac{64}{63} - \frac{27}{63}$$

$$= \frac{37}{63}$$

## Guided Practice

A continuous random variable is given by $f(x) = \frac{3x^2}{117}$ on $[2, 5]$. Find:

**a** $P(3 \leq X \leq 4)$

$\frac{37}{117}$

**b** $P(X \leq 4)$

$\frac{56}{117}$

For $f(x) = \frac{x^2}{21}$, find the value of $c$ such that $P(1 \leq X \leq c) = \frac{1}{2}$.

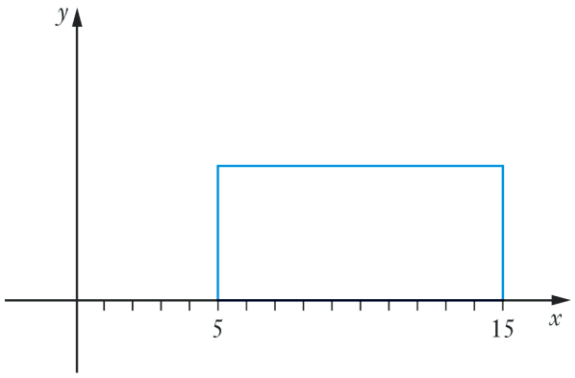
$\sqrt[3]{\frac{65}{2}}$

## Key Ideas

1. For a continuous PDF, the integral $\int_a^b f(x)\,dx$ finds $P($ ..... $)$
2. If a requested interval extends beyond the domain, the integration limits are ..... to the domain boundary.

## **Investigation** Uniform probability distribution functions

Consider a uniform continuous probability distribution in the domain $[5, 15]$.




A uniform distribution has equal probabilities throughout its domain, this means $f(x)$ is ……

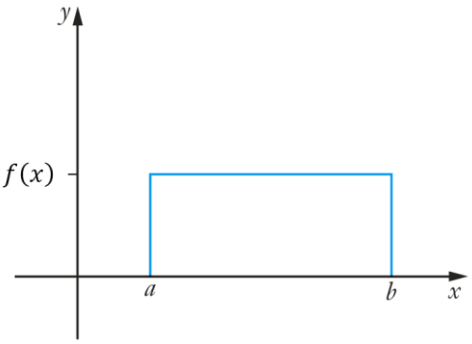
The area under the graph is therefore the shape of a ……

The area of the rectangle is ……

The width of the rectangle is ……

Therefore, the height of the rectangle is ……

So, $f(x) =$ ……




Generalising: For a uniform PDF defined for $[a, b]$,

The area of the rectangle is ……

The width of the rectangle is ……

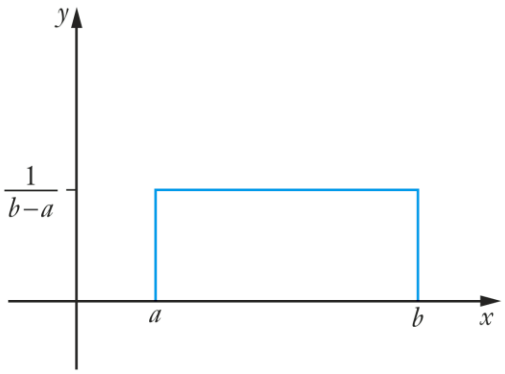
So $f(x) =$ ……

## Uniform Distributions

For a **uniformly distributed** PDF on $[a, b]$,

The PDF is

$$f(x) = \begin{cases} \frac{1}{b-a} & \text{for } a \le x \le b \\ 0 & \text{otherwise} \end{cases}$$




## Example Calculate probability for a uniform PDF

The continuous random variable $X$ is uniformly distributed on $[2, 8]$.

Find the PDF $f(x)$.

$$f(x) = \frac{1}{8-2} = \frac{1}{6}$$

Find $P(X \le 5)$.

$$\int_2^5 \frac{1}{6} dx = \left[\frac{x}{6}\right]_2^5 = \frac{5}{6} - \frac{2}{6}$$

$$= \frac{1}{2}$$

## Guided Practice

$X$ is a continuous random variable defined on $[7, 17]$.

**a** Find the PDF $f(x)$

$f(x) = \frac{1}{10}$

**b** Find $P(X > 10)$

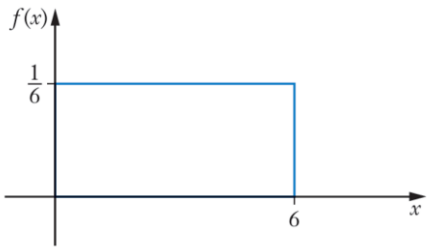
$\frac{7}{10}$

## Key Ideas

1. A uniform distribution function defined for $[a, b]$ is the c..... function:

   $f(x) =$ .....

**12** For the continuous probability distribution graphed, find:




**a** $P(X \leq 3)$

$P(X \leq 3) = \frac{1}{2}$

**b** $P(1 \leq X \leq 2)$

$P(1 \leq X \leq 2) = \frac{1}{6}$

**c** $P(1 \leq X \leq 4)$

$P(1 \leq X \leq 4) = \frac{1}{2}$

**d** $P(X < 4)$

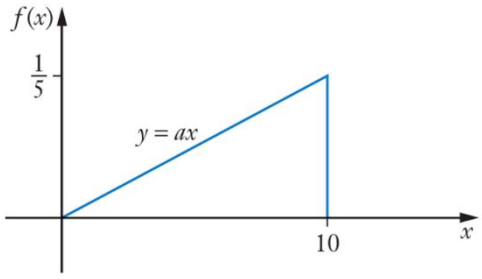
$P(X < 4) = \frac{2}{3}$

**e** $P(X \geq 4)$

$P(X \geq 4) = \frac{1}{3}$

Probability is the area under the curve, so, for a uniform distribution: $\frac{1}{6} \times$ width

**13** A PDF is shown with $f(x) = ax$.




**a** Find the equation of the linear function $y = ax$.

$\int_0^{10} (ax)\,dx = 50a = 1$

$a = \frac{1}{50}$

$y = \frac{x}{50}$

**b** Find:

**i** $P(X < 9)$

$\left[\frac{x^2}{100}\right]_0^9 = \frac{81}{100}$

**ii** $P(X \leq 3)$

$\left[\frac{x^2}{100}\right]_0^3 = \frac{9}{100}$

**iii** $P(4 \leq X \leq 7)$

$\left[\frac{x^2}{100}\right]_4^7 = \frac{33}{100}$

**iv** $P(2 < X < 6)$

$\left[\frac{x^2}{100}\right]_2^6 = \frac{8}{25}$

**v** $P(X > 5)$

$\left[\frac{x^2}{100}\right]_5^{10} = \frac{3}{4}$

**14**

**a** Show that the uniform distribution $f(x) = \frac{1}{b-a}$ for $a \leq x \leq b$, 0 for all other $x$, is a PDF.

$\text{Area} = (b-a) \times \frac{1}{b-a} = 1$

$f(x) \geq 0 \text{ on } [a, b].$

Therefore, it is a PDF.

**b** If $a = 3$ and $b = 7$, find:

**i** $P(X \leq 6)$

**ii** $P(X \geq 5)$

**iii** $P(5 \leq X \leq 6)$

b. With $a = 3, b = 7, f(x) = \frac{1}{4}$ on $[3, 7]$

i. $P(X \leq 6) = \left(\frac{1}{4}\right)(3) = \frac{3}{4}$

ii. $P(X \geq 5) = \left(\frac{1}{4}\right)(2) = \frac{1}{2}$

iii. $P(5 \leq X \leq 6) = \left(\frac{1}{4}\right)(1) = \frac{1}{4}$

**15** The continuous PDF is defined by $f(x) = ax^2$ in the domain [0, 5].

**a** Evaluate $a$.

$\int_0^5 (ax^2)\,dx = \frac{125a}{3} = 1$

$a = \frac{3}{125}$

**b** Find:

**i** $P(X \leq 3)$

$\left[\frac{x^3}{125}\right]_0^3 = \frac{27}{125}$

**ii** $P(1 < X < 4)$

$\left[\frac{x^3}{125}\right]_1^4 = \frac{63}{125}$

**iii** $P(X > 2)$

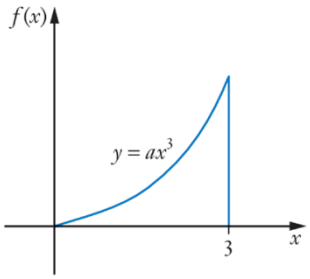
$\left[\frac{x^3}{125}\right]_2^5 = \frac{117}{125}$

**16** The continuous random variable $X$ has PDF $f(x) = ax^3$ on [0, 3].

**a** Evaluate $a$.

$\int_0^3 (ax^3)\,dx = \frac{81a}{4} = 1$

$a = \frac{4}{81}$




**b** Find:

**i** $P(1 \leq X \leq 3)$

$\left[\frac{x^4}{81}\right]_1^3 = \frac{80}{81}$

**ii** $P(X < 2)$

$\left[\frac{x^4}{81}\right]_0^2 = \frac{16}{81}$

**17** A continuous probability function is given by $f(x) = ke^x$, defined on [1, 6].

**a** Find the exact value of $k$.

$$\int_1^6 (ke^x)\,dx = k(e^6 - e) = 1$$

$$k = \frac{1}{e(e^5 - 1)}$$

**b** Find each exact probability:

**i** $P(2 \leq X \leq 5)$

$$P(2 \leq X \leq 5) = k(e^5 - e^2) = \frac{e(e^3 - 1)}{e^5 - 1}$$

**ii** $P(X < 4)$

$$P(X < 4) = k(e^4 - e) = \frac{e^3 - 1}{e^5 - 1}$$

**iii** $P(X \geq 3)$

$$P(X \geq 3) = k(e^6 - e^3) = \frac{e^2(e^3 - 1)}{e^5 - 1}$$

**18**

**a** Show that $y = \sin x$ is a PDF in the domain $\left[0, \frac{\pi}{2}\right]$.

$$\int_0^{\frac{\pi}{2}} \sin x\,dx = [-\cos x]_0^{\frac{\pi}{2}} = -\cos\left(\frac{\pi}{2}\right) + \cos(0) = 0 + 1 = 1.$$

$$\sin x \geq 0 \text{ on } \left[0, \frac{\pi}{2}\right]$$

**b** Find each exact probability:

**i** $P\left(X \leq \frac{\pi}{3}\right)$

$$P\left(X \leq \frac{\pi}{3}\right) = [-\cos x]_0^{\frac{\pi}{3}} = -\frac{1}{2} + 1 = \frac{1}{2}$$

**ii** $P\left(0 < X < \frac{\pi}{4}\right)$

$$P\left(0 < X < \frac{\pi}{4}\right) = [-\cos x]_0^{\frac{\pi}{4}} = 1 - \frac{1}{\sqrt{2}} = \frac{\sqrt{2} - 1}{\sqrt{2}}$$

**iii** $P\left(X > \frac{\pi}{6}\right)$

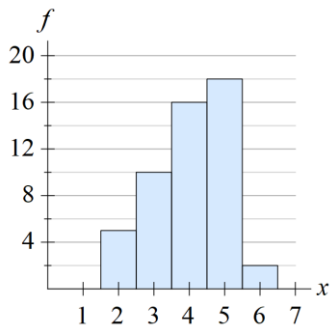
$$P\left(X > \frac{\pi}{6}\right) = [-\cos x]_{\frac{\pi}{6}}^{\frac{\pi}{2}} = 0 + \frac{\sqrt{3}}{2} = \frac{\sqrt{3}}{2}$$

# Mode of a Continuous Distribution

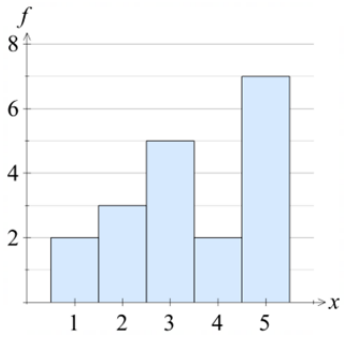
## Review

- [ ] Identify mode from a histogram

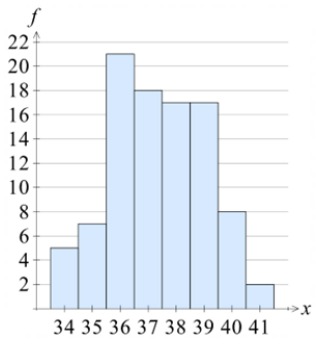
**a**




**b**




**c**




- [ ] Find global maxima

Find global minima and maxima in the domain $[0.5, 5]$ for $f(x) = x^3 - 6x^2 + 9x - 4$.
Remember to confirm the nature of the point and check the endpoints of the domain.

global max at $(5,16)$
global min at $(3,-4)$

## ⓘ Mode of a Continuous Distribution

The mode is the value of $x$ at which the PDF reaches its maximum.

To find the mode, we find the PDF's maximum turning point.

1. Differentiate $f(x)$
2. Solve $f'(x) = 0$
3. Check $x$ is in the domain of $f(x)$
4. Confirm it is the global maximum using the second derivative or a gradient table, and by checking the function value at the boundaries.

⚠ For a function that is always increasing $f'(x) > 0$, the mode is the end of its domain.

⚠ For a function that is always decreasing $f'(x) < 0$, the mode is the start of its domain.

## Example Find the mode of a continuous probability function

A continuous random variable $X$ has PDF defined on $[0,4]$: $f(x) = \frac{3x(4-x)}{32}$

Find its mode.

1. Differentiate

$$f(x) = \frac{3}{32}(4x - x^2)$$

$$f'(x) = \frac{3}{32}(4 - 2x)$$

$$f'(x) = \frac{3}{16}(2 - x)$$

2. Solve $f'(x) = 0$

$$0 = \frac{3}{16}(2 - x)$$

$$x = 2$$

3. Check domain

$x = 2$ is in $[0, 4]$ ✓

4. Check maximum

$$f''(x) = -\frac{3}{16} < 0 \therefore \max$$

$$f(0) = 0, \quad f(2) = \frac{3}{8}, \quad f(4) = 0$$

$\therefore x = 2$ is the mode.

## Guided Practice

**a** Find the mode of $f(x) = \frac{x(6-x)}{18}$ on $[0, 6]$.

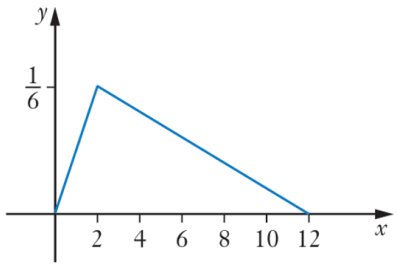
$x = 3$

**b** Find the mode of $f(x) = \frac{3x^2}{26}$ on $[1,3]$.

$x = 3$

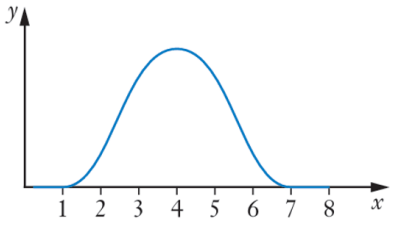
**1** Find the mode of each continuous probability distribution.

**a**



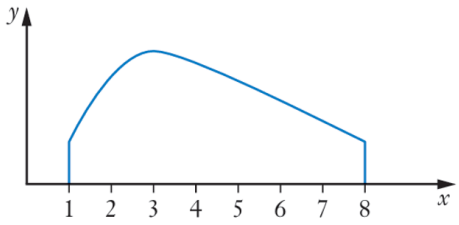

2

**b**



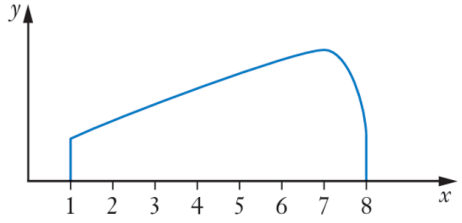

4

**c**




3

**d**




7

**2** Find the mode of each continuous probability distribution.

**a** $f(x) = -\frac{3(x^2 - 8x - 9)}{434}$ defined in the domain [0, 7]

This is a concave down parabola. The turning point must be maximum.

$$f'(x) = -\frac{3(2x - 8)}{434} = 0$$

$$x = 4$$

Mode = 4

**b** $f(x) = -\frac{3(x^2 - 16x + 15)}{1100}$ defined in the domain [1, 11]

This is a concave down parabola. The turning point must be maximum.

$$f'(x) = -\frac{3(2x - 16)}{1100} = 0$$

$$x = 8$$

Mode = 8.

**c** $f(x) = \frac{4e^{4x}}{e^{8(e^{16} - 1)}}$ defined in the interval $2 \leq x \leq 6$

$f'(x) = \frac{16e^{4x}}{e^{8(e^{16} - 1)}} > 0$ always increasing on $[2, 6]$.

The maximum is at the end of the domain.

Mode = 6.

**3** Find the mode of $f(x) = \frac{2(2x^3 - 33x^2 + 168x + 3)}{2105}$ defined in the interval $0 \leq x \leq 5$

$$f'(x) = \frac{2(6x^2 - 66x + 168)}{2105} = 0$$

$$x^2 - 11x + 28 = 0$$

$$(x - 4)(x - 7) = 0.$$

$x = 4$ is in $[0, 5]$.

Check second derivative:

$$f''(x) = 2x - 11$$

$$f''(4) = -3 < 0 \therefore \max$$

Check $f(0) = \frac{6}{2105}, f(4) = \frac{110}{421} \approx 0.26, f(5) = \frac{536}{2105} \approx 0.25$:

Mode = 4.

**4** Determine whether each function is a PDF. If it is, find its mode.

**a** $f(x) = 3x^2$, where $0 \leq x \leq 1$

$$\int_0^1 3x^2 \, dx = 1,$$

$f(x) \geq 0$ for all $x$

$f'(x) = 6x > 0$, increasing. Mode = 1.

**b** $f(x) = \frac{1}{4}x$, where $1 \leq x \leq 5$

$$\int_1^5 \frac{1}{4}x \, dx = \frac{24}{8} = 3 \neq 1$$

Not a PDF.

**c** $f(x) = \frac{4 - 2x}{3}$, where $0 \leq x \leq 3$

$$\int_0^3 \frac{4 - 2x}{3} \, dx = \left[\frac{4x}{3} - \frac{x^2}{3}\right]_0^3 = 4 - 3 = 1$$

but $f(x) < 0$ for $x > 2$. Not a PDF.

**d** $f(x) = (n + 1)x^n$, where $0 \leq x \leq 1, n \geq 0$

$$\int_0^1 (n + 1)x^n \, dx = [x^{n+1}]_0^1 = 1$$

$f(x) \geq 0$ for all positive $x$

$f'(x) = n(n + 1)x^{n-1} \geq 0$, increasing.

Mode = 1.

**e** $f(x) = \frac{1}{2}\sin x$, where $0 \leq x \leq \pi$

$$\int_0^\pi \frac{1}{2}\sin x \, dx = \frac{1}{2}[-\cos x]_0^\pi = 1$$

$f(x) \geq 0$

Mode at max value of sine: $\sin x = 1$: $x = \frac{\pi}{2}$.

# Cumulative Distribution Functions

## Review

☐ Integrate to a variable upper limit

**a**

$$\int_1^x t^2\,dt$$

**b**

$$\int_0^x 3t\,dt$$

**c**

$$\int_2^x t^3\,dt$$

☐ Find cumulative frequency for a frequency table

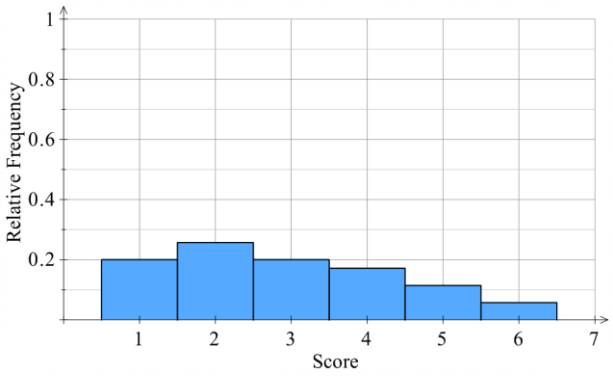
**a**

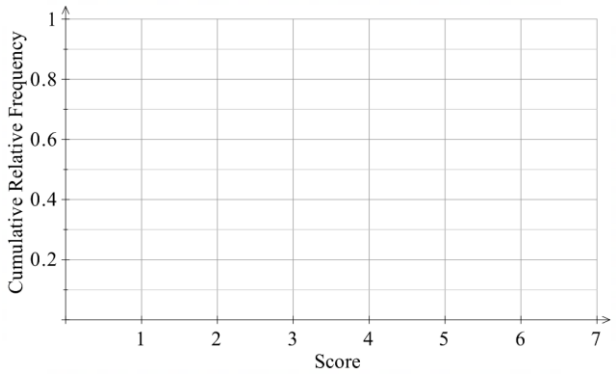
| Score | Frequency | Cumulative frequency |
|---|---|---|
| 4 | 4 | |
| 5 | 6 | |
| 6 | 7 | |
| 7 | 10 | |
| 8 | 5 | |

**b**

| Number of calls | Frequency | Cumulative frequency |
|---|---|---|
| 20 | 4 | |
| 21 | 3 | |
| 22 | 10 | |
| 23 | 12 | |
| 24 | 6 | |
| 25 | 5 | |

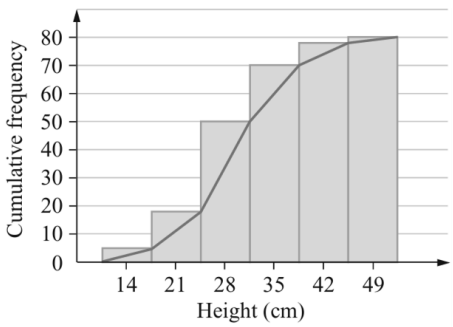
☐ Construct cumulative frequency histogram from regular histogram



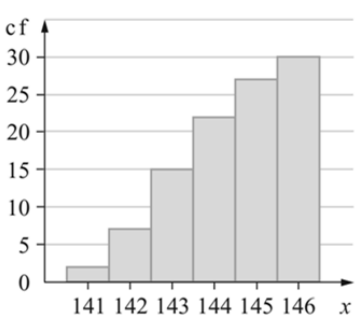




☐ Find the median and IQR from a cumulative frequency polygon

**a**




**b**

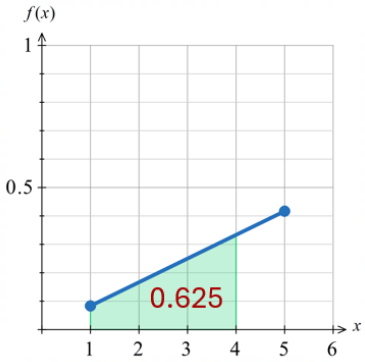


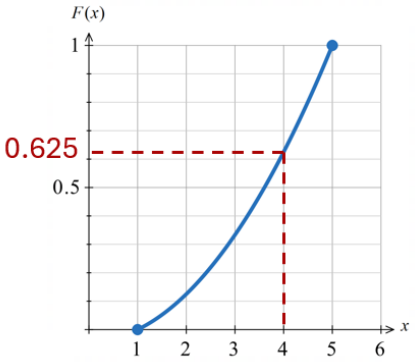
## ⓘ Cumulative Distribution Function

For a continuous probability distribution $f(t)$ defined on $[a, b]$:

The **cumulative distribution function (CDF)**, $F(x)$, gives the probability that $X$ is less than or equal to $x$. It is the area under the graph from $a$ to $x$, so it is the integral from $a$ to $x$.

$$F(x) = P(X \leq x) = \int_a^x f(t)\,dt$$




**Properties:**

- The PDF is the derivative of the CDF.

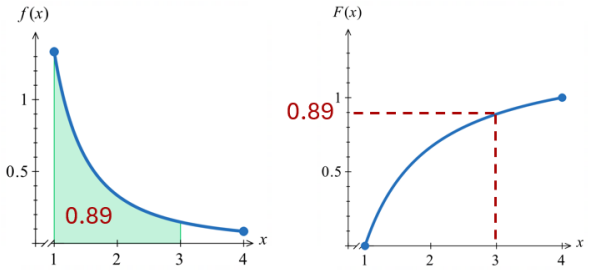
- A CDF is **always monotonically increasing**.
  The cumulative probability cannot decrease.

- At the left boundary of the domain, $F(a) = 0$.
- At the right boundary of the domain, $F(b) = 1$.
- For any $x$ between $a$ and $b$, $0 < F(x) < 1$.

## ⓘ Properties of the CDF

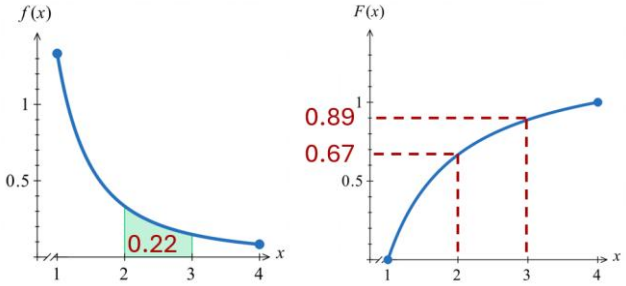
- The function value of the CDF gives the cumulative probability.

$$F(a) = P(X \leq a)$$



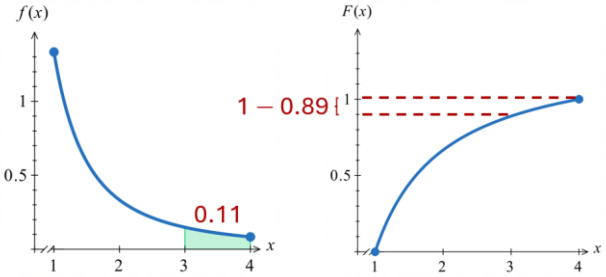

- To find the probability of an interval using the CDF, subtract the CDF values at the two endpoints.

$$P(c \leq X \leq d) = F(d) - F(c)$$




- To find the probability of $X$ greater than or equal to $a$, subtract the probability less than or equal to $a$ from 1.

$$P(X \geq a) = 1 - P(X \leq a)$$
$$= 1 - F(a)$$

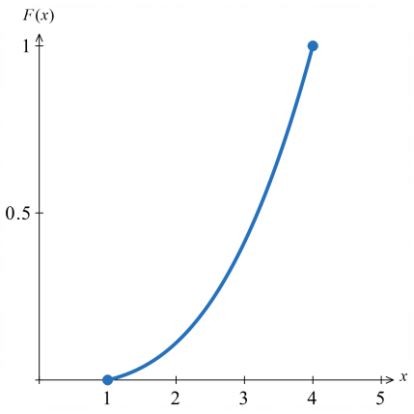



## Example Find the cumulative distribution function

The continuous random variable has PDF $f(x) = \frac{x^2}{21}$ defined on $[1, 4]$.

Find the CDF and use it to find $P(X \leq 3)$ and $P(2 \leq x \leq 3)$.

$$F(x) = \int_1^x \frac{t^2}{21} dt$$
$$= \frac{1}{21}\left[\frac{t^3}{3}\right]_1^x$$
$$= \frac{1}{21}\left(\frac{x^3}{3} - \frac{1}{3}\right)$$
$$= \frac{x^3 - 1}{63}$$




$$P(X \leq 3) = F(3) = \frac{26}{63}$$

$$P(2 \leq X \leq 3) = F(3) - F(2) = \frac{26}{63} - \frac{7}{63} = \frac{19}{63}$$

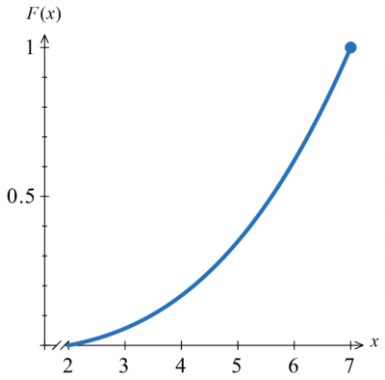
## Guided Practice

A PDF is $f(x) = \frac{3x^2}{335}$ on $[2,7]$.

**a** Find the cumulative distribution function.

$\frac{x^3 - 8}{335}$

The CDF has been sketched for you:




**b** Find $P(X \leq 4)$

$\frac{56}{335}$

**c** $P(3.5 \leq X \leq 6.2)$

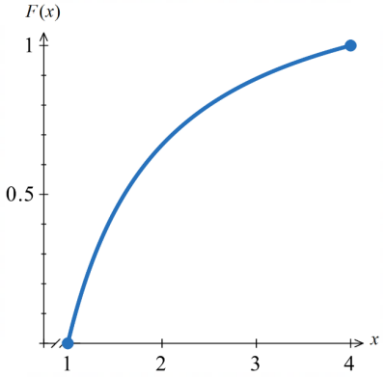
0.583

A PDF is $f(x) = \frac{4}{3x^2}$ on $[1,4]$.

**a** Find the cumulative distribution function.

$\frac{4}{3} - \frac{4}{3x}$

The CDF has been sketched for you:




**b** Find $P(2 \leq X \leq 3)$

$\frac{2}{9}$

**c** $P(X \geq 2)$

$\frac{1}{3}$

**1** Find the cumulative distribution function for each continuous probability distribution.

**a** $f(x) = \frac{x^2}{9}$ defined in the domain [0, 3]

$$F(x) = \int_0^x \left(\frac{t^2}{9}\right) dt = \frac{x^3}{27}$$

**b** $f(x) = \frac{e^x}{e^4 - 1}$ in the interval $0 \leq x \leq 4$

$$F(x) = \int_0^x \left(\frac{e^t}{e^4 - 1}\right) dt = \frac{e^x - 1}{e^4 - 1}$$

**c** $f(x) = \frac{4(x-2)^3}{625}$ in the domain [2, 7]

$$F(x) = \int_2^x \left(\frac{4(t-2)^3}{625}\right) dt = \frac{(x-2)^4}{625}$$

**d** $f(x) = \frac{3x(8-x)}{135}$ in the domain [2, 5]

$$F(x) = \int_2^x \left(\frac{3t(8-t)}{135}\right) dt = \left(\frac{3}{135}\right)\left[4t^2 - \frac{t^3}{3}\right]_2^x$$
$$= \frac{12x^2 - x^3 - 40}{135}$$

**2** For each PDF, find $F(x)$ and confirm $F(b) = 1$.

**a** $f(x) = \frac{1}{32}x$, where $0 \leq x \leq 8$

$$F(x) = \int_0^x \frac{t}{32}\, dt = \frac{x^2}{64}.$$
$$F(8) = \frac{64}{64} = 1$$

**b** $f(x) = \frac{3}{16}x^2$, where $-2 \leq x \leq 2$

$$F(x) = \int_{-2}^x \frac{3t^2}{16}\, dt = \left[\frac{t^3}{16}\right]_{-2}^x = \frac{x^3 + 8}{16}.$$
$$F(2) = \frac{16}{16} = 1$$

**c** $f(x) = \frac{3}{2}(1 - x^2)$, where $0 \leq x \leq 1$

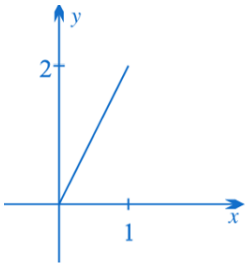
$$F(x) = \int_0^x \frac{3}{2}(1 - t^2)\, dt = \frac{3x - x^3}{2}$$
$$F(1) = \frac{2}{2} = 1$$

**d** $f(x) = \frac{1}{e}(e^x + 1)$, where $0 \leq x \leq 1$

$$F(x) = \frac{1}{e}\int_0^x (e^t + 1)\, dt = \frac{1}{e}(e^x + x - 1)$$
$$F(1) = \frac{1}{e}(e) = 1$$

**3** The function $y = 2x, 0 \leq x \leq 1$, is a PDF.

**a** Sketch the graph and verify $f(x) \geq 0$.




$f(x) = 2x \geq 0$ for $x \geq 0$,

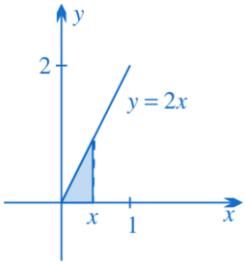
so $f(x) \geq 0$ in $[0,1]$

**b** Show the area bounded by the function and the $x$-axis is 1; confirm by integration.

Triangle with base 1 and height 2: area = $\frac{1}{2}(1)(2) = 1$

$\int_0^1 2x\,dx = [x^2]_0^1 = 1$

**c** Use area formulae to show the CDF is $P(X \leq x) = x^2$.




Area is half base times height

$A = \frac{1}{2}x \times 2x = x^2$

**d** Confirm by calculating $\int_0^x 2t\,dt$.

$[t^2]_0^x = x^2$

**e** Use the CDF to find $Q_1$, $Q_2$ and $Q_3$.

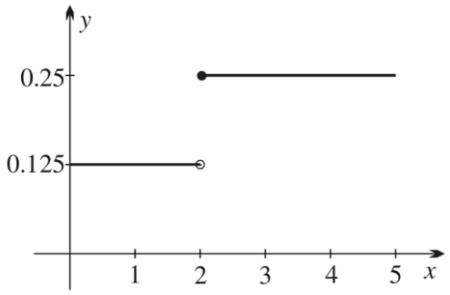
$Q1$: $\frac{1}{2}$

$Q2$: $\frac{1}{\sqrt{2}}$

$Q3$: $\frac{\sqrt{3}}{2}$

**4** A piecewise function is graphed: $f(x) = 0.125$ for $0 \leq x < 2, f(x) = 0.25$ for $2 \leq x \leq 5$.

**a** Verify it forms a valid PDF.




Area $= 2(0.125) + 3(0.25) = 0.25 + 0.75 = 1$.

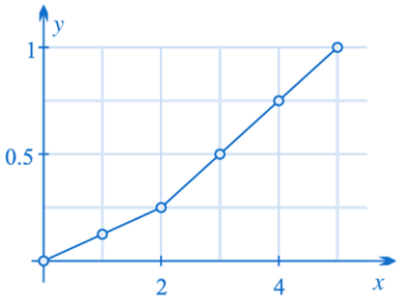
$f(x) \geq 0$. Valid PDF.

**b** Complete the cumulative probability table $P(X \leq x)$ for $x = 0, 1, 2, 3, 4, 5$.

| $x$ | 0 | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|---|
| $P(X \leq x)$ | | | | | | |

$0, \frac{1}{8}, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 1$

**c** Plot and graph the CDF and write the CDF in piecewise notation.

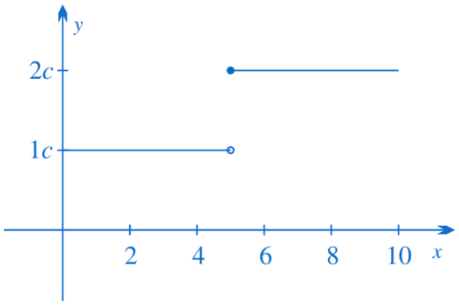



$$F(x) = \begin{cases} \frac{1}{8}x, & \text{for } 0 \leq x < 2 \\ \frac{1}{4}x - \frac{1}{4}, & \text{for } 2 \leq x \leq 5 \end{cases}$$

**5** A PDF is defined by

$$f(x) = \begin{cases} c, & \text{for } 0 \leq x \leq 5 \\ 2c, & \text{for } 5 < x \leq 10 \end{cases}$$

**a** Sketch the PDF.




**b** Find $c$.

Area $= 5c + 5(2c) = 15c = 1$

$c = \frac{1}{15}$

**c** Find the CDF.

$$F(x) = \begin{cases} cx = \frac{x}{15}, & \text{for } 0 \leq x < 5 \\ 2cx - 5c = \frac{2x}{15} - \frac{1}{3}, & \text{for } 5 \leq x \leq 10 \end{cases}$$

**d** Use the CDF to find $P(1 < X < 7)$.

$P(1 < X < 7) = F(7) - F(1)$

$= \left(\frac{14}{15} - \frac{1}{3}\right) - \frac{1}{15}$

$= \frac{8}{15}$

**6** A continuous probability distribution is defined by $f(x) = \frac{2e^{2x}}{e^{10}-1}$ in the domain [0, 5].

**a** Find the CDF.

$$F(x) = \int_0^x \left(\frac{2e^{2t}}{e^{10}-1}\right) dt = \frac{e^{2x}-1}{e^{10}-1}$$

**b** Calculate each probability correct to 2 significant figures.

**i** $P(X \leq 2)$

$$P(X \leq 2) = \mathrm{F}(2) = \frac{e^4-1}{e^{10}-1} \approx 0.0024$$

**ii** $P(X \leq 4)$

$$P(X \leq 4) = \frac{e^8-1}{e^{10}-1} \approx 0.14$$

**iii** $P(X > 3)$

$$P(X > 3) = 1 - \frac{e^6-1}{e^{10}-1} \approx 0.98$$

**iv** $P(X \geq 2.8)$

$$P(X \geq 2.8) = 1 - F(2.8) \approx 0.99$$

**v** $P(2 \leq X \leq 4)$

$$P(2 \leq X \leq 4) = F(4) - F(2) \approx 0.13$$

**7**

**a** Find the exact value of $a$ if $f(x) = \frac{a}{x}$ is a continuous PDF defined in the domain [1, 6].

$$\int_1^6 \frac{a}{x} dx = a(\ln 6 - \ln 1) = 1$$

$$a = \frac{1}{\ln 6}.$$

**b** Find the CDF.

$$F(\mathrm{x}) = \int_1^x \left(\frac{1}{t \cdot \ln 6}\right) dt = \frac{\ln x}{\ln 6}$$

**c** Find to 2 decimal places:

**i** $P(X \leq 3)$

$$P(X \leq 3) = \frac{\ln 3}{\ln 6} \approx 0.61$$

**ii** $P(X \leq 2)$

$$P(X \leq 2) = \frac{\ln 2}{\ln 6} \approx 0.39$$

**iii** $P(X > 5)$

$$P(X > 5) = 1 - \frac{\ln 5}{\ln 6} \approx 0.10$$

**iv** $P(X \geq 4)$

$$P(X \geq 4) = 1 - \frac{\ln 4}{\ln 6} \approx 0.23$$

**v** $P(2 \leq X \leq 5)$

$$P(2 \leq X \leq 5) = \frac{\ln 5 - \ln 2}{\ln 6} = \frac{\ln\left(\frac{5}{2}\right)}{\ln 6} \approx 0.51$$

**8**

**a** Show that $y = \cos x$ is a PDF in the domain $\left[\frac{3\pi}{2}, 2\pi\right]$.

$$\int_{\frac{3\pi}{2}}^{2\pi} \cos x \ dx = [\sin x]_{\frac{3\pi}{2}}^{2\pi}$$

$$= \sin(2\pi) - \sin\left(\frac{3\pi}{2}\right)$$

$$= 0 - (-1) = 1$$

$\cos x \geq 0$ on $\left[\frac{3\pi}{2}, 2\pi\right]$ (cos positive in Q4)

Therefore, it is a PDF

**b** Find the CDF.

$$F(x) = \int_{\frac{3\pi}{2}}^{x} \cos t \ dt$$

$$= [\sin t]_{\frac{3\pi}{2}}^{x}$$

$$= \sin x + 1$$

**c** Find each probability in exact form:

**i** $P\left(X \leq \frac{5\pi}{3}\right)$

$$P\left(X \leq \frac{5\pi}{3}\right) = \sin\left(\frac{5\pi}{3}\right) + 1$$

$$= -\frac{\sqrt{3}}{2} + 1$$

$$= \frac{2 - \sqrt{3}}{2}$$

**ii** $P\left(X \geq \frac{7\pi}{4}\right)$

$$P\left(X \geq \frac{7\pi}{4}\right) = 1 - F\left(\frac{7\pi}{4}\right)$$

$$= 1 - \left(\sin\left(\frac{7\pi}{4}\right) + 1\right)$$

$$= -\sin\left(\frac{7\pi}{4}\right)$$

$$= \frac{1}{\sqrt{2}}$$

**iii** $P\left(\frac{5\pi}{3} \leq X \leq \frac{11\pi}{6}\right)$

$$P\left(\frac{5\pi}{3} \leq X \leq \frac{11\pi}{6}\right) = F\left(\frac{11\pi}{6}\right) - F\left(\frac{5\pi}{3}\right)$$

$$= \sin\left(\frac{11\pi}{6}\right) - \sin\left(\frac{5\pi}{3}\right)$$

$$= -\frac{1}{2} - \left(-\frac{\sqrt{3}}{2}\right)$$

$$= \frac{\sqrt{3} - 1}{2}$$

**9**

**a** Find the CDF for $f(x) = \frac{5x^4}{7776}$ for $1 \leq x \leq 6$.

$$F(x) = \int_1^x \left(\frac{5t^4}{7776}\right) dt = \frac{x^5 - 1}{7776}$$

**b** Find:

**i** $P(X \leq 3)$

$$P(X \leq 3) = F(3) = \frac{121}{3888}$$

**ii** $P(X \leq 2)$

$$P(X \leq 2) = F(2) = \frac{31}{7776}$$

**iii** $P(X < 5)$

$$P(X < 5) = F(5) = \frac{781}{1944}$$

**iv** $P(X > 4)$

$$P(X > 4) = 1 - F(4)$$

$$= 1 - \frac{1023}{7776} = \frac{2251}{2592}$$

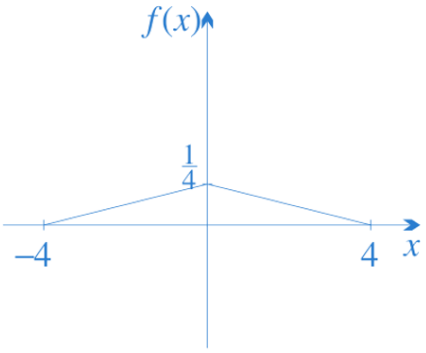
**v** $P(2 \leq X \leq 4)$

$$P(2 \leq X \leq 4) = F(4) - F(2) = \frac{31}{243}$$

**10** A probability density function is defined piecewise by

$$f(x) = \begin{cases} k(4+x) & -4 \le x \le 0 \\ k(4-x) & 0 \le x \le 4 \end{cases}$$

**a** Find the value of the constant $k$. Hence write the equation of $f(x)$ and sketch it.




$\int_{-4}^{4} f(x)\,dx = 1$. Split region based on piecewise definition:

$$\int_{-4}^{0} k(4+x)\,dx + \int_{0}^{4} k(4-x)\,dx$$

$$= k\left[4x + \left(\frac{1}{2}\right)x^2\right]_{-4}^{0} + k\left[4x - \left(\frac{1}{2}\right)x^2\right]_{0}^{4}$$

$$= k\big((0+0) - (-16+8)\big) + k\big((16-8) - (0-0)\big)$$

$$= 8k + 8k = 16k = 1$$

$$k = \frac{1}{16}$$

$$f(x) = \begin{cases} \frac{1}{16}(4+x) & -4 \le x \le 0 \\ \frac{1}{16}(4-x) & 0 \le x \le 4 \end{cases}$$

**b** What is the probability that $0 \le X \le 2$?

$$P(0 \le X \le 2) = \int_{0}^{2} \left(\frac{1}{16}\right)(4-x)\,dx$$

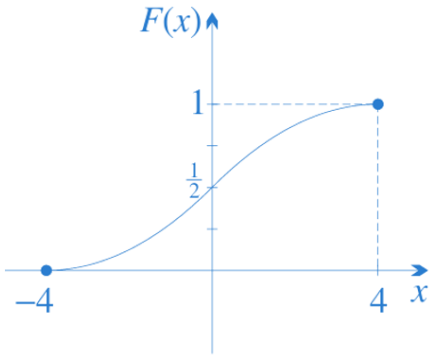
$$= \left(\frac{1}{16}\right)\left[4x - \left(\frac{1}{2}\right)x^2\right]_{0}^{2}$$

$$= \frac{1}{16}\big((8-2) - 0\big) = \frac{3}{8}$$

**c** Why is the median zero, and what is the mode?

$f$ is symmetric about $x = 0$, so equal area lies either side and the median = 0.
The mode is $x = 0$ (global maximum of $f$).

**d** Find the CDF for $-4 \le x \le 0$, and the CDF for $0 \le x \le 4$. Then sketch the whole CDF.




For $-4 \le x \le 0$:

$$F(x) = \int_{-4}^{x} \frac{1}{16}(4+t)\,dt$$

$$= \frac{1}{32}[(4+t)^2]_{-4}^{x}$$

$$= \frac{1}{32}\big((4+x)^2 - 0\big) = \frac{1}{32}(4+x)^2$$

For $0 \le x \le 4$, note $F(0) = \frac{1}{2}$:

$$F(x) = \frac{1}{2} + \int_{0}^{x} \frac{1}{16}(4-t)\,dt$$

$$= \frac{1}{2} - \frac{1}{32}[(4-t)^2]_{0}^{x}$$

$$= \frac{1}{2} - \frac{1}{32}\big((4-x)^2 - 16\big) = 1 - \frac{1}{32}(4-x)^2$$

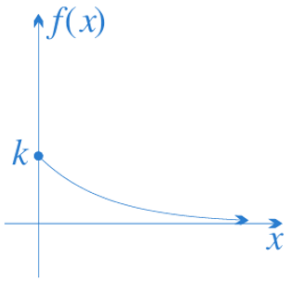
**11** Let $f(x)$ and $F(x) = e^{-kx}$, where $k = \frac{1}{8}\ln 2$, be the PDF and CDF respectively for the experiment of observing the time $x$ days that a radioactive nucleus survives before decaying.

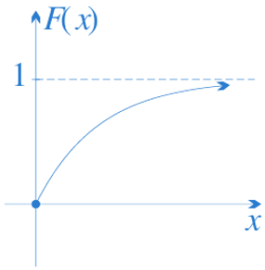
**a** Explain why the domain of possible values is the unbounded interval $[0, \infty)$.

**b** Find the formula for the PDF, and sketch the CDF and PDF.

**c** Find the median, and show it is the half-life.

**d** Find the probabilities that it decays on the first day and after the first day.




a) Decay can theoretically occur at any time, no matter how large, so the domain [0, ∞) is used.

b) CDF: $F(x) = 1 - e^{-kx}$

Differentiating: $f(x) = ke^{-kx}$

c) Set $F(x) = 0.5$:

$1 - e^{-kx} = \frac{1}{2}$

$e^{-kx} = \frac{1}{2}$

$kx = \ln 2$

$x = \frac{\ln 2}{k} = \frac{\ln 2}{\left(\frac{1}{8}\right)\ln 2}$ = 8 days, which is the half-life.

d) $P(X \leq 1) = F(1) = 1 - e^{-k} \approx 0.083$

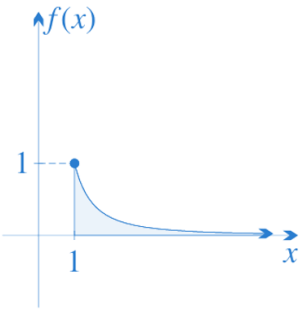
$P(X > 1) = 1 - F(1) = e^{-k} \approx 0.917$

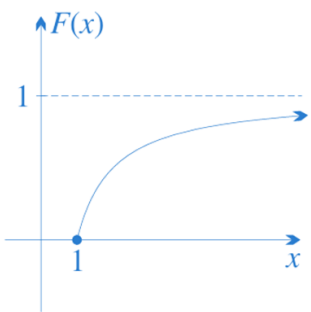
**12**

**a** Find the shaded area under the curve $y = \frac{1}{x^2}$ over the interval $(1, \infty)$.

**b** Hence show that $y = \frac{1}{x^2}$ for $x \geq 1$, is a PDF, and find the CDF.

**c** Sketch the PDF and CDF on two axes.




a) $\int_1^\infty \left(\frac{1}{x^2}\right) dx = \left[-\frac{1}{x}\right]_1^\infty = 0 - (-1) = 1$

(The limit of $-\frac{1}{x}$ as $x \to \infty$ is 0.)

b) $f(x) = \frac{1}{x^2} \geq 0$ for all $x \geq 1$, and the integral over $[1, \infty)$ equals 1, so it is a PDF.

$F(x) = \int_1^x \left(\frac{1}{t^2}\right) dt = \left[-\frac{1}{t}\right]_1^x = 1 - \frac{1}{x}$

**19**

**a** Find the mode of $f(x) = -\frac{3}{22}(x^2 - 6x + 5)$ defined on [2, 4].

This is a concave down parabola, the turning point is the maximum

$$f'(x) = -\frac{3}{22}(2x - 6) = 0$$

$$x = 3$$

Mode = 3.

**b** Find the CDF.

$$F(x) = \int_2^x \left(-\frac{3}{22}(t^2 - 6t + 5)\right) dt$$

$$= -\frac{3}{22}\left[\frac{t^3}{3} - 3t^2 + 5t\right]_2^x$$

$$= -\frac{1}{22}(x^3 - 9x^2 + 15x - 2)$$

**c** Find $P(X \leq a)$ where $a$ is the mode.

$$P(X \leq 3) = F(3) = -\frac{1}{22}(27 - 81 + 45 - 2)$$

$$= -\frac{1}{22}(-11) = \frac{1}{2}$$

**20** Athletes' finishing times varied between 3 and 7 minutes, represented by the continuous PDF $f(x) = \frac{1}{116}(x^3 - 9x^2 + 24x + 1)$ on [3, 7].

**a** Find the CDF.

$$F(x) = \int_3^x \left(\frac{1}{116}(t^3 - 9t^2 + 24t + 1)\right) dt$$

$$= \frac{1}{464}(x^4 - 12x^3 + 48x^2 + 4x - 201)$$

**b** Find the probability that an athlete finishes:

**i** in less than 5 minutes

$$P(X < 5) = F(5) = \frac{625 - 1500 + 1200 + 20 - 201}{464} = \frac{144}{464} = \frac{9}{29}$$

**ii** in 4 minutes or more

$$P(X \geq 4) = 1 - F(4) = 1 - \frac{256 - 768 + 768 + 16 - 201}{464} = 1 - \frac{71}{464} = \frac{393}{464}$$

**iii** in between 4 and 5 minutes

$$P(4 \leq X \leq 5) = F(5) - F(4) = \frac{144}{464} - \frac{71}{464} = \frac{73}{464}$$

**c** What is the most likely finishing time?

f'(x) = (1/116)(3x² − 18x + 24) = 0 → x² − 6x + 8 = 0 → x = 2 or x = 4. Only x = 4 is in [3, 7].

Check endpoints: f(3) = 19/116, f(4) = 17/116, f(7) = 71/116. Maximum at x = 7.

Mode = 7 minutes.

**13** A chicken wanders uniformly at random inside a circular enclosure of radius 6 m.
Let $X$ be its distance from the centre O.

**a** Find the CDF, which is the probability of finding the chicken a distance of $x$ metres or less from the centre of the enclosure. $F(x) = P(X \leq x)$.

$$\mathrm{F(x)} = \frac{\text{area of inner circle}}{\text{area of whole circle}}$$

$$= \frac{\pi x^2}{\pi \times 6^2}$$

$$= \frac{1}{36}x^2, \text{for } 0 \leq \mathrm{x} \leq 6$$

**b** Find the probability of finding the chicken 2 metres or less from the centre of the enclosure.

$$F(2) = \frac{1}{36}(4) = \frac{1}{9}$$

**c** Find the probability of finding the chicken between two to five metres from the centre of the circular enclosure.

$$P(\text{chook is } 2-5 \text{ m from centre}) = F(5) - F(2)$$

$$= \left(\frac{1}{36}\right)(25 - 4)$$

$$= \frac{21}{25}$$

**d** Use the CDF to find the probability of finding the chicken within 6 m of the centre of the enclosure.

$$F(6) = \frac{1}{36}(6)^2 = 1$$

Based on the context, this makes sense as the enclosure only has a radius of 6 m

**e** Find the PDF.

PDF is the derivative of the CDF

$$f(x) = \frac{x}{18}$$

# Median and Quartiles using the CDF

## Median and Quartiles using the CDF

The median is the value of $x$ for which $F(x) = 0.5$.

The first quartile can be similarly found by $F(x) = 0.25$ and the third quartile $F(x) = 0.75$.

## Example Calculate median and quartiles using a CDF

The continuous random variable $X$ has CDF $F(x) = \frac{x^3-1}{63}$ on $[1,4]$.

**a** Find the median

$$F(x) = \frac{1}{2}$$
$$\frac{x^3-1}{63} = \frac{1}{2}$$
$$x^3 - 1 = \frac{63}{2}$$
$$x^3 = \frac{65}{2}$$
$$x = \sqrt[3]{\frac{65}{2}} \approx 3.19$$

**b** Find the first quartile $Q_1$

$$F(x) = \frac{1}{4}$$
$$\frac{x^3-1}{63} = \frac{1}{4}$$
$$x^3 - 1 = \frac{63}{4}$$
$$x^3 = \frac{67}{4}$$
$$x = \sqrt[3]{\frac{67}{4}} \approx 2.56$$

## Guided Practice

A continuous probability distribution has PDF $f(x) = \frac{x^4}{11605}$ in the domain $[4,9]$.

Find the CDF and then find the median and third quartile.

Q2: 7.86
Q3: 8.51

## Key Ideas

1. The median can be found from the ..... ..... ..... by setting it equal to 0.5

**1** For the CDF $F(x) = \frac{x^2}{64}$, find

**a** the median $Q_2$

**b** $Q_1$ and $Q_3$

$Q_2 = 4\sqrt{2}$

$Q_1$: 4

$Q_3$: $4\sqrt{3}$.

**2** For the CDF $F(x) = \frac{x^3+8}{16}$, find

**a** the median $Q_2$

**b** $Q_1$ and $Q_3$

$Q_2 = 0$

$Q_1$: $-\sqrt[3]{4}$

$Q_3$: $\sqrt[3]{4}$.

**3** Find the median of each continuous random variable correct to 2 decimal places.

**a** $f(x) = \frac{3x^2}{511}$, interval $1 \leq x \leq 8$

$F(x) = \frac{x^3 - 1}{511}$

$\frac{m^3 - 1}{511} = \frac{1}{2}$

$m^3 = 256.5$

$m \approx 6.35$

**b** $f(x) = \frac{3(x-3)^2}{16}$, domain [1, 5]

$F(x) = \frac{(x-3)^3 + 8}{16}$

$(m-3)^3 = 0$

$m = 3$

**c** $f(x) = \frac{(2x-1)^4}{16105}$, domain [1, 6]

$F(x) = \frac{(2x-1)^5 - 1}{161050}$

$(2m-1)^5 = 80526$

$2m - 1 \approx 9.58$

$m \approx 5.29$

**d** $f(x) = \frac{x(x^2-3)^3}{3570}$, domain [2, 4]

$F(x) = \frac{(x^2-3)^4 - 1}{28560}$

$(m^2-3)^4 = 14281$

$m^2 - 3 \approx 10.93$

$m \approx 3.73$

**4** For $f(x) = \frac{x^2}{168}$ defined on $2 \leq x \leq 8$, find:

**a** the median

$$F(x) = \frac{x^3 - 8}{504} = 0.5$$
$$m^3 = 260$$
$$m \approx 6.38$$

**b** the 1st quartile

$$F(x) = \frac{x^3 - 8}{504} = 0.25$$
$$m^3 = 134$$
$$m \approx 5.12$$

**c** the 3rd quartile

$$F(x) = \frac{x^3 - 8}{504} = 0.75$$
$$m^3 = 386$$
$$m \approx 7.28$$

**d** the 67th **percentile**; 67% probability to get a value this number or less.

$$F(x) = \frac{x^3 - 8}{504} = 0.67$$
$$m^3 = 345.68$$
$$m \approx 7.02$$

**e** the 8th **decile**; 80% probability to get a value this number or less.

$$F(x) = \frac{x^3 - 8}{504} = 0.8$$
$$m^3 = 411.2$$
$$m \approx 7.44$$

**5** For $f(x) = \frac{x^2}{576}$ on $0 \leq x \leq 12$, find to 2 d.p.

**a** the 20th percentile

**b** the median

**c** the 3rd quartile

$$F(x) = \frac{x^3}{1728}$$

20th percentile: 7.02

Median: 9.52

3rd quartile: 10.90

**6** Define $f(x) = \frac{3}{32}x(4 - x), 0 \leq x \leq 4$.

**a** State the mode.

Mode = 2. Graph is a concave down parabola with maximum at $x = 2$.

**b** Confirm $\int_0^4 f(x)\,dx = 1$.

$$\frac{3}{32}\int_0^4 x\,(4-x)\,dx = \frac{3}{32}\left[2x^2 - \frac{x^3}{3}\right]_0^4$$

$$= \frac{3}{32}\left(32 - \frac{64}{3}\right)$$

$$= 1$$

**c** Write down $P(X \leq 2)$. What property of the curve allows this without integration?

$P(X \leq 2) = \frac{1}{2}$ by symmetry about $x = 2$.

**d** Evaluate $P(X \leq 1)$ and $P(X > 1)$. Explain why they add to 1.

$$P(X \leq 1) = \frac{3}{32}\left[2x^2 - \frac{x^3}{3}\right]_0^1 = \frac{5}{32}$$

$$P(X > 1) = 1 - \frac{5}{32} = \frac{27}{32}$$

They add to 1 because total probability is 1.

**e** Evaluate $P(X \leq 0.5)$ and $P(X \geq 3.5)$. What do you notice?

$$P(X \leq 0.5) = \frac{3}{32}\left[2(0.25) - \frac{0.125}{3}\right] = \frac{3}{32}\left(\frac{11}{24}\right) = \frac{11}{256}$$

$P(X \geq 3.5) = \frac{11}{256}$ by symmetry.

They are equal.

**f** Find the CDF using $F(x) = \int_0^x f(t)\,dt$.

$$F(x) = \int_0^x \frac{3}{32}t(4-t)\,dt = \frac{3}{32}\left[2t^2 - \frac{t^3}{3}\right]_0^x$$

$$= \frac{x^2(6-x)}{32}$$

**g** Use the CDF to evaluate:

**i** $P(X < 1.5)$

**ii** $P(1 < X < 1.5)$

**iii** $P(3 < X < 3.5)$

**iv** $P(2 < X < 2.5)$

i) $F(1.5) = \frac{(2.25)(4.5)}{32} = \frac{81}{256}$

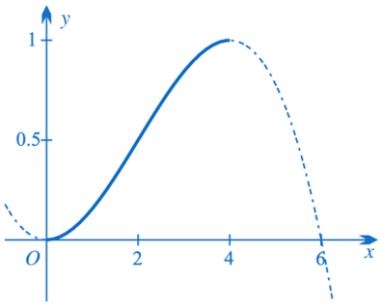
ii) $F(1.5) - F(1) = \frac{81}{256} - \frac{40}{256} = \frac{41}{256}$

iii) By symmetry = $P(0.5 < X < 1) = F(1) - F(0.5)$

$$= \frac{40}{256} - \frac{11}{256} = \frac{29}{256}$$

iv) $F(2.5) - F(2) = \frac{(6.25)(3.5)}{32} - \frac{16}{32} = \frac{47}{256}$

**h** Graph the CDF.




**i** Confirm $P(X < 2) = 0.5$ using the CDF.

$$F(2) = \frac{4(4)}{32} = \frac{16}{32} = \frac{1}{2}$$

50% of data lie to the left of $x = 2$.

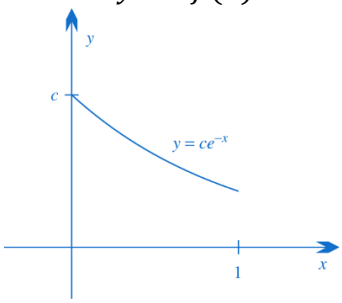
**j** Find $Q_1$ and $Q_3$ by solving $F(x) = 0.25$ and $F(x) = 0.75$.

$$Q_1 \approx 1.3$$

$$Q_3 \approx 2.7$$

**7** Define $f(x) = ce^{-x}, 0 \leq x \leq 1$.

**a** Sketch $y = f(x)$.




**b** Find $c$, given $f(x)$ is a PDF.

$$\int_0^1 c\,e^{-x}\,dx = c(1 - e^{-1}) = \frac{c(e-1)}{e} = 1$$

$$c = \frac{e}{e-1}$$

**c** Find the CDF.

$$F(x) = \frac{e}{e-1}\int_0^x e^{-t}\,dt = \frac{e}{e-1}(1 - e^{-x})$$

**d** Find a general equation for finding any percentile by setting $F(x) = p$

Set $F(x) = p$

$$e^{-x} = 1 - \frac{p(e-1)}{e}$$

$$x = \log\left(\frac{e}{e - p(e-1)}\right)$$

**e** Hence find $Q_1$, $Q_2$ and $Q_3$.

$$Q_1 : \left(p = \frac{1}{4}\right) : x = \ln\left(\frac{4e}{3e+1}\right)$$

$$Q_2 : \left(p = \frac{1}{2}\right) : x = \ln\left(\frac{2e}{e+1}\right)$$

$$Q_3 : \left(p = \frac{3}{4}\right) : x = \ln\left(\frac{4e}{e+3}\right)$$

# Expected Value and Variance

## Review

Calculate expected value and variance from a discrete random distribution

The table below shows the monetary return for playing a game of chance at a casino.

Find the expected value $E(X) = \Sigma x P(x)$

| $x$ | $-\$10$ | $\$10$ | $\$90$ |
|---|---|---|---|
| $P(x)$ | $\frac{3}{4}$ | $\frac{3}{13}$ | $\frac{1}{52}$ |

$-\$3.46$

Find the variance $\mathrm{Var}(X) = \mathrm{E}(X^2) - \mu^2 = \sum\left(x^2 P(x)\right) - \mu^2$

and standard deviation $\sigma = \sqrt{\mathrm{VAR}(X)}$

| $x$ | $-\$10$ | $\$10$ | $\$90$ |
|---|---|---|---|
| $P(x)$ | $\frac{3}{4}$ | $\frac{3}{13}$ | $\frac{1}{52}$ |

$\$241.86$
$\sigma = 15.55$

## Expected Value and Variance of Continuous Random Variables

### Expected Value (mean)

Discrete:

$$E(X) = \mu = \sum xP(x)$$

Continuous:

$$E(X) = \mu = \int_a^b xf(x)dx$$

### Variance

**Discrete**:

$$\text{Var}(X) = E(X^2) - \mu^2 = \sum \left(x^2P(x)\right) - \mu^2$$

**Continuous**:

$$\text{Var}(X) = E(X^2) - \mu^2 = \int_a^b x^2f(x)dx - \mu^2$$

## Example Calculate expected value and variance for a PDF

The continuous random variable $X$ has PDF $f(x) = \frac{3x^2}{8}$ on $[0,2]$.

Find the expected value.

$$E(X) = \int_0^2 x \cdot \frac{3x^2}{8}dx$$
$$= \int_0^2 \frac{3x^3}{8}dx$$
$$= \left[\frac{3x^4}{32}\right]_0^2$$
$$= \frac{3}{32}(16 - 0)$$
$$= \frac{3}{2}$$

Find the variance.

$$\text{Var}(X) = \int_0^2 x^2 \cdot \frac{3x^2}{8}dx - \mu^2$$
$$= \int_0^2 \frac{3x^4}{8}dx - \left(\frac{3}{2}\right)^2$$
$$= \left[\frac{3x^5}{40}\right]_0^2 - \frac{9}{4}$$
$$= \frac{3}{40}(32 - 0) - \frac{9}{4}$$
$$= \frac{3}{20}$$

## Guided Practice

The continuous random variable $X$ has PDF $f(x) = \frac{x^2}{21}$ on $[1,4]$. Find $E(X)$ and $\text{Var}(X)$.

$E(X) = \frac{85}{28}, \text{Var}(X) = \frac{2067}{3920}$

**1** A function is defined by $f(x) = \frac{1}{10}, 0 \leq x \leq 10$.

**a** Show $f(x)$ is a valid PDF.

$f(x) = \frac{1}{10} \geq 0$

$\int_0^{10} \frac{1}{10}\, dx = 1$

**b** Calculate $E(X)$ using $E(X) = \int_a^b xf(x)\, dx$.

$E(X) = \int_0^{10} \frac{x}{10}\, dx = \frac{1}{10}\left[\frac{x^2}{2}\right]_0^{10}$

$= 5$

**c** Does $E(X)$ agree with your understanding of expected value as an average?

$E(X) = 5$ is the midpoint of [0,10], the expected average for a uniform distribution

**d** Calculate $\mathrm{Var}(X) = E(X^2) - \left(E(X)\right)^2$ then find $\sigma$ using $\sigma = \sqrt{\mathrm{Var}}$

$E(X^2) = \frac{1}{10}\int_0^{10} x^2\, dx = \frac{100}{3}$

$\mathrm{Var} = \frac{100}{3} - 25 = \frac{25}{3}$

$\sigma = \frac{5}{\sqrt{3}}$

**2** Define $f(x) = \frac{3}{2}x^2$, domain $[-1, 1]$.

**a** Confirm it is a valid PDF.

$\int_{-1}^{1} \frac{3}{2}x^2\, dx = \frac{3}{2}\left[\frac{x^3}{3}\right]_{-1}^{1} = \frac{3}{2}\left(\frac{2}{3}\right) = 1$

$f(x) \geq 0$ as $\frac{3}{2}x^2 \geq 0$

**b** Find E(X).

$E(X) = \frac{3}{2}\int_{-1}^{1} x^3\, dx = 0$

**c** Find $\mathrm{Var}(X)$ and $\sigma$.

$E(X^2) = \frac{3}{2}\int_{-1}^{1} x^4\, dx = \frac{3}{2}\left(\frac{2}{5}\right) = \frac{3}{5}$

$\mathrm{Var}(X) = \frac{3}{5} - 0 = \frac{3}{5}$

$\sigma = \frac{\sqrt{15}}{5}$

**d** Calculate $\int_{\mu-\sigma}^{\mu+\sigma} f(x)\, dx$ to find the percentage of the population within one standard deviation of the mean.

$P\left(-\frac{\sqrt{15}}{5} \leq X \leq \frac{\sqrt{15}}{5}\right) = \frac{3}{2}\int_{-\frac{\sqrt{15}}{5}}^{\frac{\sqrt{15}}{5}} x^2\, dx = 3\left[\frac{x^3}{3}\right]_0^{\frac{\sqrt{15}}{5}}$

$= \left(\frac{\sqrt{15}}{5}\right)^3 \approx 0.46$

so approximately 46%

**3** Find the percentage of the population within one standard deviation of the mean, to 2 s.f:

**a** $f(x) = 2x$, domain [0, 1]

**b** $f(x) = |x|$, domain [−1, 1]

**c** $f(x) = \frac{3}{64}x^2$, domain [0, 4]

a) 63%

b) 50%

c) 67%